

EE202L - Final Project - Corrupted Chord

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Introduction:

In this project, we designed and implemented an active notch filter which suppressed the fundamental frequency of an unfretted G string. We stored this active notch filter in the cigar-box surrounding the neck. The system combines mechanical guitar construction with analog circuit design, using a twin-T bandstop filter augmented with positive feedback to achieve a narrow bandwidth and deep attenuation at the target frequency. Component values were selected from a restricted set and verified through analytical calculations and SPICE simulations before being implemented on a printed circuit board. The completed circuit was integrated with the guitar electronics and experimentally tuned to account for real-world component tolerances. Final testing evaluates the filter's frequency response and its effectiveness in selectively reducing the G-string fundamental without significantly affecting adjacent pitches.

We selected a "standard" design.

Filter Design

We define the nodes:

- Node 1: Between the two resistors R and the capacitor 2C.
- Node 2: Between the two capacitors C and the resistor R/2.
- Node Out: Output voltage V_{out} .
- Node x: Reference node V_x , connected to GND

KCL at Node 1:

$$(V_1 - V_{in})/R + (V_1 - V_{out})/R + v1(s*2C) = 0$$

Solve for V_1 :

$$V_1(2/R + 2*s*C) = (V_{in} + V_{out})/R$$

$$V_1 = (V_{in} + V_{out}) / (2(1 + sRC))$$

KCL at Node 2:

$$(V_2 - V_{in})sC + (V_2 - V_{out})sC + v2/(R/2) = 0$$

Solve for V_2 :

$$V_2(2sC + 2/R) = sC(V_{in} + V_{out})$$

$$V_2 = sRC(V_{in} + V_{out}) / (2(1 + sRC))$$

KCL at the output node V_{out} :

$$(V_{out} - V_1)/R + (V_{out} - V_2)sC = 0$$

$$V_{out}(1 + sRC) = V_1 + sRC*V_2$$

Substitute output equation:

$$V_{out}(1 + sRC) = [(V_1 + V_{out}) / (2(1 + sRC))] + sRC * [sRC(V_{in} + V_{out}) / (2(1 + sRC))]$$

Multiply by $2(1 + sRC)$ to clear the denominator:

$$2V_{out}(1 + sRC)^2 = (V_{in} + V_{out}) + (sRC)^2 * (V_{in} + V_{out})$$

$$2V_{out}(1 + 2sRC + s^2 R^2 C^2) = (V_{in} + V_{out})(1 + s^2 R^2 C^2)$$

$$V_{out} [2 + 4sRC + 2s^2 R^2 C^2 - 1 - s^2 R^2 C^2] = V_{in}(1 + s^2 R^2 C^2)$$

$$V_{out} [s^2 R^2 C^2 + 4sRC + 1] = V_{in}(s^2 R^2 C^2 + 1)$$

$$V_{out} / V_{in} = (s^2 R^2 C^2 + 1) / (s^2 R^2 C^2 + 4sRC + 1)$$

$$V_{out} / V_{in} = (s^2 + 1/(RC)^2) / (s^2 + s(4/RC) + 1/(RC)^2)$$

$$H(s) = K(s^2 + \omega^2) / (s^2 + s(\omega/Q) + \omega^2)$$

We can equate the terms:

$$\omega^2 = 1/(RC)^2 \rightarrow \omega = 1/RC$$

The coefficient of s in the denominator is ω / Q .

$$4/RC = \omega / Q$$

Substitute $\omega = 1/RC$:

$$4 * \omega = \omega / Q$$

$$4 = 1/Q$$

$$Q = 1/4$$

The gain K is 1.

Therefore, $\omega = 1/RC$, $Q = 1/4$, and $K = 1$.

Active (Bootstrapped) Filter Derivation

We now modify the circuit using positive feedback such that the node voltage v_x is equal to αv_{out} , where $\alpha < 1$.

KCL with $v_x = \alpha * v_{out}$ instead of $v_x = 0$.

KCL at Node 1 (connected to v_x via $2C$):

$$(V_1 - V_{in})/R + (V_1 - V_{out})/R + (V_1 - V_x)(s2C) = 0$$

Substitute $V_x = \alpha * V_{out}$:

$$(V_1 - V_{in})/R + (V_1 - V_{out})/R + (V_1 - \alpha * V_{out})(s2C) = 0$$

$$V_1(2/R + 2sC) = (V_{in} + V_{out})/R + 2*s*C*\alpha*V_{out}$$

$$V_1 = [V_{in} + V_{out}(1 + 2*\alpha*s*R*C)] / [2(1 + s*R*C)]$$

KCL at Node 2 (connected to V_x via $R/2$):

$$(V_2 - V_{in})*s*C + (V_2 - V_{out})*s*C + (V_2 - V_x)/(R/2) = 0$$

Substitute $V_x = \alpha * V_{out}$:

$$V_2(2*s*C + 2/R) = s*C*(V_{in} + V_{out}) + (2*\alpha*V_{out})/R$$

$$V_2 = [s*R*C*V_{in} + V_{out}(s*R*C + 2*\alpha)] / [2(1 + s*R*C)]$$

KCL at Output Node (remains the same relative to v_1 and v_2):

$$V_{out}(1 + s*R*C) = V_1 + s*R*C*V_2$$

Substitute into the output equation. Let $T = RC$ for simplicity:

$$V_{out}(1 + sT) = [V_{in} + V_{out}(1 + 2*\alpha*sT)] / [2(1 + sT)] + sT * [sT*V_{in} + V_{out}(sT + 2*\alpha)] / [2(1 + sT)]$$

Multiply by $2(1 + sT)$:

$$2V_{out}(1 + sT)^2 = V_{in} + V_{out}(1 + 2*\alpha*sT) + s^2T^2V_{in} + sT*V_{out}(sT + 2*\alpha)$$

Expand and group terms:

$$\text{Left Side: } V_{out}(2 + 4*sT + 2s^2T^2)$$

$$\text{Right Side: } V_{in}(1 + s^2T^2) + V_{out}(1 + 2*\alpha*sT + s^2T^2 + 2*\alpha*sT)$$

$$\text{Right Side simplified: } V_{in}(1 + s^2T^2) + V_{out}(1 + s^2T^2 + 4*\alpha*sT)$$

Move V_{out} terms to the left:

$$V_{out} [(2*s^2T^2 + 4*sT + 2) - (s^2T^2 + 4*\alpha*sT + 1)] = V_{in}(1 + s^2T^2)$$

$$V_{out} [s^2T^2 + sT(4 - 4\alpha) + 1] = V_{in}(1 + s^2T^2)$$

Solve for the transfer function:

$$V_{out} / V_{in} = (s^2 T^2 + 1) / (s^2 T^2 + s T^4 (1 - \alpha) + 1)$$

Divide by T^2 (where $T = RC$):

$$V_{out} / V_{in} = (s^2 + 1/(RC)^2) / (s^2 + s * [4(1 - \alpha)/RC] + 1/(RC)^2)$$

Comparing this to the standard form:

The constant term remains $1/(RC)^2$, so ω is unchanged.

The s coefficient is now:

$$\omega / Q_{new} = 4(1 - \alpha) * (1/RC)$$

$$\omega = 1/RC$$

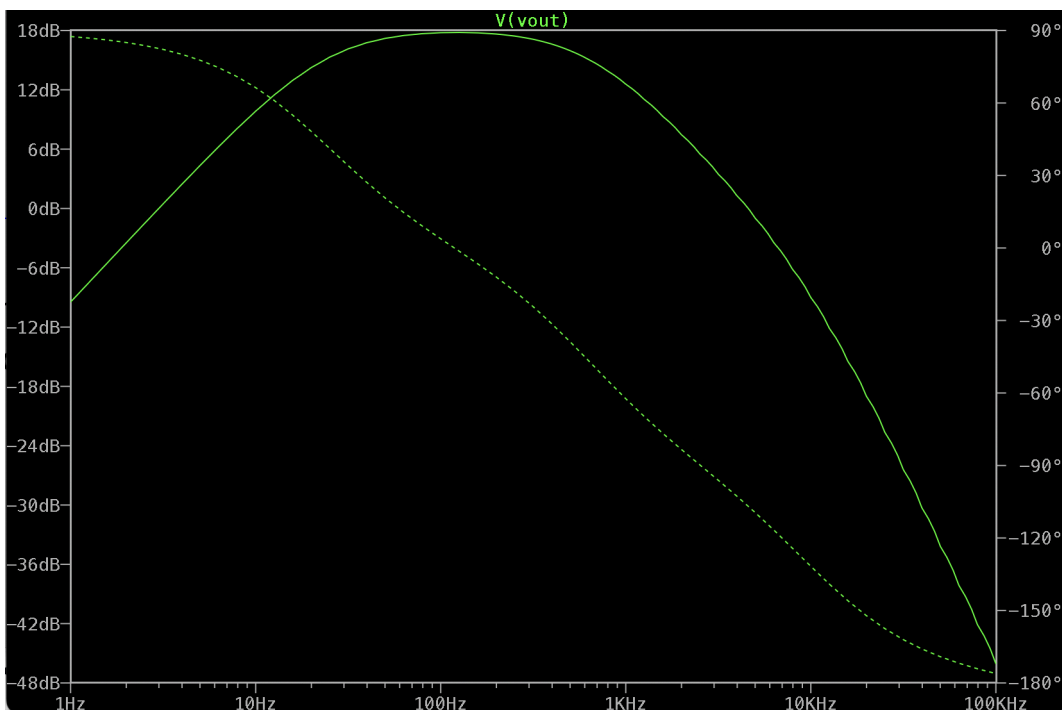
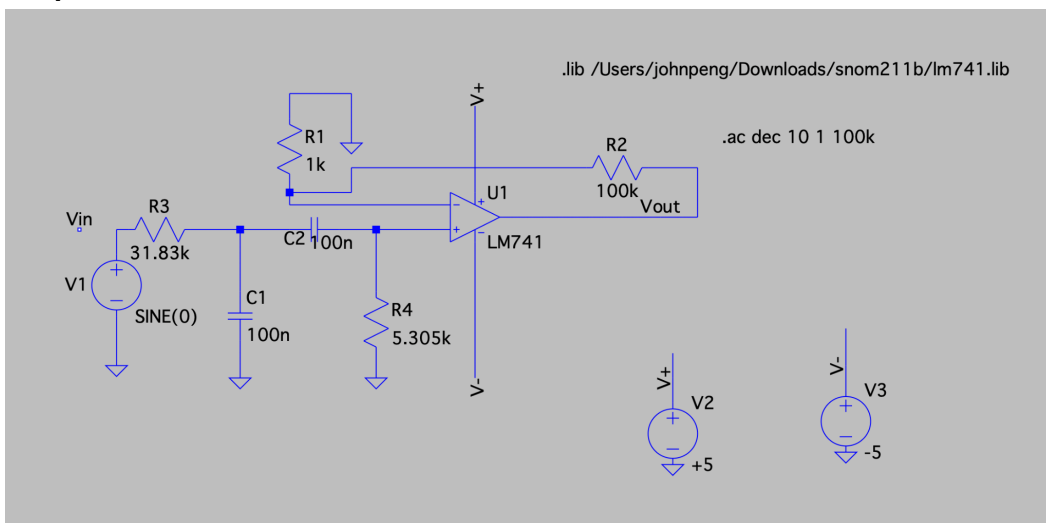
$$\omega / Q_{new} = 4(1 - \alpha) * \omega$$

$$1 / Q_{new} = 4(1 - \alpha)$$

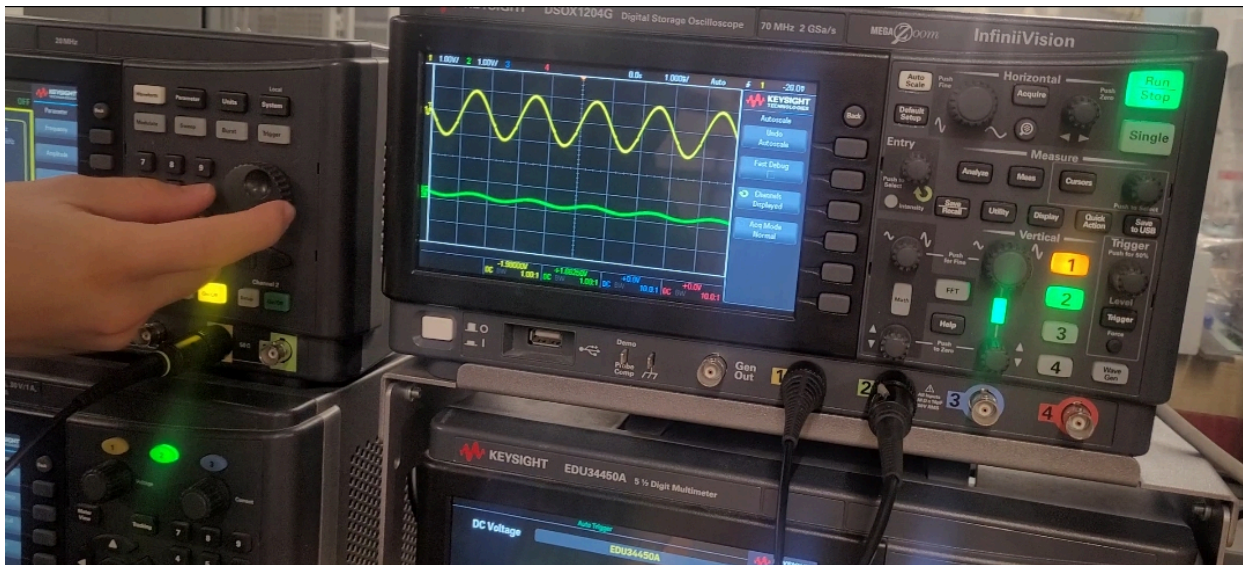
$$Q_{new} = 1 / [4(1 - \alpha)]$$

Therefore, the center frequency ω remains constant, the quality factor Q is increased by $1/(1-\alpha)$

LTSpice Simulation:



Oscilloscope Testing:



Future Improvements: Luckily we ran into few errors during this project. Our sole problem was the filter not fully filtering some lower frequencies. To combat this, we only had to increase our potentiometer readings slightly, as they were slightly off our target values. Other than this, there were no technical issues with the amplification and filtering PCB; however, there are some methods we could have implemented better.

When getting the physical components we struggled with identifying placements as we had mislabeled values in KiCAD, better documentation would have saved us from this. Additionally, we would take more care with soldering as during our testing process we had identified a poor connection for the 9V input. Had we tested each connection as we went through the soldering process, we would have noticed this sooner and not spent extra time chasing the connection.

Conclusions:

In this project, we designed, simulated, and implemented an active twin-T notch filter intended to suppress the 196 Hz fundamental frequency of an unfretted G string in a cigar-box electric guitar while preserving adjacent frequencies. Derivations and LTSpice simulations confirmed that introducing positive feedback increased the quality factor without shifting the notch frequency, allowing for attenuation at the target pitch. During physical testing, the primary challenge encountered was incomplete attenuation of lower frequencies, which was resolved through fine adjustment of potentiometer values to better align the real-world circuit with the theoretical design. Aside from minor assembly and documentation issues, the PCB and filtering stages functioned as intended after tuning. Overall, this project reinforced the importance of accounting for component tolerances, validating designs through simulation, and iteratively tuning analog circuits in hardware to achieve the desired performance.

John worked on the hot-plate soldering, PCB design, schematic, and woodwork.

Peter worked on the hand-soldering, testing, and wiring of bluebox.

We both worked on the calculations, component selection, glueing, guitar strings, tuning, and writeup.